

MCCONNELL AFB, KS, USA

WMO: 724505

Lat: 37.617N

Lon: 97.267W

Elev: 1371

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	12.8	-2.5	5.1	13.5	1.6	6.3	17.5	30.2	36.9	27.4	39.4	10.8	0	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	99.6	73.0	96.7	73.8	93.2	73.7	78.0	90.2	76.8	89.1	75.6	87.6	13.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.9	137.8	82.9	73.3	130.2	82.0	72.3	125.9	81.4	42.4	90.1	41.1	89.5	40.0	87.7	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.6	25.0	22.6	1.7	102.8	4.7	3.3	-1.7	105.2	-4.4	107.1	-7.1	108.9	-10.5	111.3		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			1.7	102.8	4.7	3.3	-1.7	105.2	-4.4	107.1	-7.1	108.9	-10.5	111.3		
			2.7	80.4	6.3	2.3	-1.9	82.1	-5.6	83.4	-9.2	84.7	-13.8	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	33.9	38.2	48.0	56.8	67.1	76.8	80.8	79.8	72.3	59.7	47.1	36.7
	DBStd	18.84	10.94	11.65	11.57	9.15	8.03	6.23	5.66	5.77	7.81	9.10	9.77	9.78
	HDD50	1688	507	356	174	40	1	0	0	0	1	23	164	422
	HDD65	4247	966	751	533	270	75	3	0	0	24	210	539	876
	CDD50	4681	7	25	112	244	531	804	955	923	668	325	77	10
	CDD65	1766	0	0	7	24	139	357	490	458	242	47	2	0
	CDH74	18689	0	2	62	245	1203	3719	5699	5047	2301	397	14	0
	CDH80	8716	0	0	9	43	393	1703	2936	2540	991	100	1	0
Wind	WSAvg	11.1	11.0	11.5	12.8	13.0	11.4	11.2	9.9	9.6	10.2	10.8	11.4	10.8
Precipitation	PrecAvg	36.1	0.8	1.4	2.5	3.3	5.5	5.9	4.2	4.1	2.9	2.9	1.5	1.1
	PrecMax	52.2	2.6	4.1	5.6	7.7	11.6	12.2	9.4	9.6	10.4	8.1	5.3	3.4
	PrecMin	25.4	0.0	0.1	0.5	0.6	2.1	2.0	0.5	0.5	0.3	0.6	0.0	0.1
	PrecStd	7.4	0.7	1.0	1.4	1.7	2.6	2.7	2.2	2.3	2.5	1.8	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	72.4	80.0	84.7	92.8	100.1	103.5	104.2	99.1	87.7	75.5	66.1
		MCWB	52.6	55.1	61.2	65.0	69.9	73.7	72.8	72.1	70.6	67.6	58.8	54.3
	2%	DB	61.0	67.0	74.8	80.5	88.1	95.9	99.6	99.6	93.0	82.4	71.6	61.2
		MCWB	48.8	52.7	59.7	63.1	69.9	74.3	73.8	72.5	71.5	64.5	58.7	51.0
	5%	DB	55.5	61.3	70.5	76.7	84.4	92.5	96.7	95.7	89.9	78.7	66.5	56.0
		MCWB	45.2	50.2	57.7	61.3	69.0	74.1	74.4	73.6	70.9	63.6	55.4	47.6
	10%	DB	50.4	56.2	66.1	72.5	81.0	89.4	93.0	91.7	86.1	74.2	62.5	51.8
		MCWB	42.4	47.3	55.6	59.7	68.3	73.2	74.4	73.5	69.4	62.5	53.4	44.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	60.0	67.7	69.4	75.4	79.2	80.1	79.3	76.3	72.6	64.3	60.1
		MCDB	61.2	64.6	72.7	76.7	84.1	91.3	91.1	91.5	89.3	79.9	69.7	63.4
	2%	WB	50.7	55.7	64.1	67.1	73.4	77.4	78.0	77.5	74.1	70.1	60.4	53.9
		MCDB	58.1	62.4	70.3	75.0	82.3	89.6	90.3	90.0	86.5	76.8	66.9	57.5
	5%	WB	46.4	52.1	60.2	65.0	71.9	76.1	76.8	76.2	72.8	67.4	57.8	48.4
		MCDB	54.0	60.5	67.6	72.9	80.5	87.6	89.4	88.5	84.5	74.0	64.3	54.3
	10%	WB	42.9	47.3	56.6	62.1	70.3	74.7	75.6	74.9	71.4	64.4	54.7	44.5
		MCDB	50.1	56.5	64.3	70.5	78.4	85.6	88.0	86.9	82.2	71.3	61.5	50.6
Mean Daily Temperature Range	5% DB	MDBR	19.7	20.5	21.4	21.0	19.7	19.8	20.2	20.5	20.8	21.0	20.3	18.3
		MCDBR	27.6	28.2	27.3	26.3	23.2	23.5	24.5	25.5	24.5	25.5	25.7	25.0
		MCWBR	17.4	17.5	15.3	14.6	10.2	8.4	6.9	7.4	8.3	12.1	14.8	16.5
	5% WB	MCDBR	25.3	24.9	22.0	22.1	19.8	21.0	20.8	20.8	21.1	19.5	21.3	22.4
		MCWBR	17.4	17.6	14.8	14.5	10.8	9.2	8.4	8.4	8.5	11.9	15.2	17.3
Clear-Sky Solar Irradiance	taub		0.287	0.298	0.333	0.373	0.408	0.413	0.421	0.425	0.380	0.337	0.298	0.280
	taud		2.498	2.477	2.408	2.319	2.296	2.312	2.329	2.272	2.374	2.484	2.524	2.526
	Ebn at Noon		288	298	295	287	278	275	272	268	276	279	281	283
	Edh at Noon		25	29	34	40	42	41	40	41	35	29	24	23
All-Sky Solar Radiation	RadAvg		800	1031	1346	1690	1891	2160	2129	1908	1587	1176	867	685
	RadStd		81	148	90	113	167	160	130	107	116	133	88	66

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page